

1. Django Project Setup

Problem:

Write a command to create a new Django project named `school_mgmt` and start a new app named `students`.

Input Format:

Project name and App name

Output Format:

Django project and app should be created.

Example 1:

Input:

`school_mgmt students`

Output:

Project '`school_mgmt`' created

App '`students`' created

=>

```
django-admin startproject school_mgmt
```

```
cd school_mgmt
```

```
python manage.py startapp students
```

2. Django Model Creation

Problem:

Create a Django model `Student` with fields: `name` (`CharField`), `age` (`IntegerField`), `email` (`EmailField`).

Input Format:

Model fields

Output Format:

Generated Django model

Example 1 Input:

`Student, name:char, age:int, email:email`

Output:

```
class Student(models.Model):
```

```
    name = models.CharField(max_length=100)
```

```
age = models.IntegerField()
email = models.EmailField()
```

=>

```
from django.db import models

class Student(models.Model):

    name = models.CharField(max_length=100)
    age = models.IntegerField()
    email = models.EmailField()

    def __str__(self):
        return self.name
```

3. Django URL Mapping

Problem:

Write the urls.py entry to map the URL /students/ to the view student_list.

Input Format:

URL path and view function name

Output Format:

Correct URL mapping in Django

Example 1 Input:

/students/ student_list

Output:

```
path('students/', views.student_list, name='student_list')
```

=>

```
from django.urls import path

from . import views

urlpatterns = [
    path('students/', views.student_list, name='student_list'),
]
```

```
students/urls.py

from django.contrib import admin
from django.urls import path, include
urlpatterns = [
    path('admin/', admin.site.urls),
    path("", include('students.urls')),
]
```

4. Django View and Template

Problem:

Write a view `student_list` that passes a list of student names to the template `students.html`.

Input Format:

List of names

Output Format:

Render template with student names

Example 1 Input:

```
["Ali", "Rahul", "Sara"]
```

Output:

Template rendered with names: Ali, Rahul, Sara

=>

```
from django.shortcuts import render

def student_list(request):
    students = ["Ali", "Rahul", "Sara"]
    return render(request, "students.html", {"students": students})
```

students.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Student List</title>
```

```
</head>
<body>
  <h2>Student Names</h2>
  <ul>
    {% for name in students %}
      <li>{{ name }}</li>
    {% endfor %}
  </ul>
</body>
</html>
```

5. Django Form Validation

Problem:

Create a Django form for Student with validation ensuring age ≥ 18 .

Input Format:

Name, Age

Output Format:

Form validation message

Example 1 Input:

John, 16

Output:

ValidationError: Age must be at least 18

Example 2 Input:

Alice, 20

Output:

Form is valid

=>

```
from django import forms
from .models import Student
class StudentForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = ['name', 'age', 'email']
    def clean_age(self):
        age = self.cleaned_data.get('age')
        if age < 18:
            raise forms.ValidationError("Age must be at least 18")
        return age
```

forms.py

```
from django.shortcuts import render
from .forms import StudentForm
def add_student(request):
    if request.method == "POST":
        form = StudentForm(request.POST)
        if form.is_valid():
            form.save()
            return render(request, "success.html")
    else:
        form = StudentForm()
    return render(request, "add_student.html", {"form": form})
```

6. Django Authentication

Problem:

Write a Django view for user login using `authenticate()` and `login()`.

Input Format:

Username, Password

Output Format:

Login success/failure

Example 1 Input:

(username=admin, password=wrongpass)

Output:

Invalid credentials

Example 2 Input:

(username=admin, password=admin123)

Output:

Login successful

=>

```
from django.shortcuts import render, redirect
```

```
from django.contrib.auth import authenticate, login
```

```
from django.contrib import messages
```

```
def user_login(request):
```

```
    if request.method == "POST":
```

```
        username = request.POST.get("username")
```

```
        password = request.POST.get("password")
```

```
        user = authenticate(request, username=username, password=password)
```

```
        if user is not None:
```

```
            login(request, user)
```

```
            messages.success(request, "Login successful")
```

```
            return redirect("home") # redirect to homepage or dashboard
```

```
        else:
```

```
            messages.error(request, "Invalid credentials")
```

```
        return render(request, "login.html")
    return render(request, "login.html")
```

login.html

```
<!DOCTYPE html>

<html>

<head>

    <title>Login</title>

</head>

<body>

    <h2>User Login</h2>

    <form method="post">

        {% csrf_token %}

        <label>Username:</label>

        <input type="text" name="username"><br><br>

        <label>Password:</label>

        <input type="password" name="password"><br><br>

        <button type="submit">Login</button>

    </form>

    {% for message in messages %}

        <p>{{ message }}</p>

    {% endfor %}

</body>

</html>
```

7. MySQL DDL Operation

Problem:

Write an SQL query to create a table courses with id, course_name, credits.

Input Format:

Table name and fields

Output Format:

DDL SQL query

Example Input:

courses id:int, course_name:varchar(100), credits:int

Output:

```
CREATE TABLE courses (  
    id INT PRIMARY KEY,  
    course_name VARCHAR(100),  
    credits INT  
);
```

=>

```
CREATE TABLE courses (  
    id INT PRIMARY KEY,  
    course_name VARCHAR(100),  
    credits INT  
);
```

8. MySQL DML Operation

Problem:

Write an SQL query to insert a new course record into the courses table.

Input Format:

Course details

Output Format:

SQL Insert statement

Example Input:

(1, "Mathematics", 4)

Output:

```
INSERT INTO courses VALUES (1, 'Mathematics', 4);
```

=>

```
INSERT INTO courses VALUES (1, 'Mathematics', 4);
```

inserting by column names -

```
INSERT INTO courses (id, course_name, credits)
```

```
VALUES (1, 'Mathematics', 4);
```

9. Flask Routing

Problem:

Create a Flask route `/hello/<name>` that returns `"Hello, <name>!"`.

Input Format:

Name in URL

Output Format:

Greeting string

Example 1 Input:

`/hello/Ali`

Output:

Hello, Ali!

Example 2 Input:

`/hello/Sara`

Output:

Hello, Sara!

=>

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route('/hello/<name>')
```

```
def hello(name):
```

```
    return f'Hello, {name}!'
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```

10. Flask Blueprint

Problem:

Create a Flask blueprint `students_bp` with a route `/students` returning "Student List Page".

Input Format:

Blueprint name and route

Output Format:

Blueprint route response

Example 1 Input:

`students_bp /students`

Output:

Student List Page

=>

```
from flask import Blueprint
```

```
students_bp = Blueprint('students_bp', __name__)
```

```
@students_bp.route('/students')
```

```
def student_list():
```

```
    return "Student List Page"
```

app.py

```
from flask import Flask
```

```
from students import students_bp
```

```
app = Flask(__name__)
```

```
app.register_blueprint(students_bp)
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```